Area Under Curve

Mind map

Curve and axis
- $\int_a^b |y|\,dx$
- Split at sign changes
- y-axis: $\int |x|\,dy$

Two curves
- $\int |f-g|\,dx$
- Find intersections first
- Split where curves cross

Strips
- Vertical: upper − lower
- Horizontal: right − left
- Choose fewer splits

Graphs to know
- Polynomials, conics
- Trig and ITF
- $e^x, \ln x, [x], \{x\}, |x|$

Sketching
- Transformations of f
- Sign of f', extrema
- Asymptotes, loops

Standard results
- $\frac{16ab}{3}$: two parabolas
- $\frac{8a^2}{3m^3}$: parabola, $y=mx$
- $\frac{|a|(\beta-\alpha)^3}{6}$
- Ellipse πab

Inequalities
- Draw boundaries
- Test a point, shade
- Common region

Shifting origin
- Area unchanged
- Shift axes and limits too
- Then standard results

Images, inverse
- Reflection keeps area
- f^{-1}: integrate f in y-strip
- f, f^{-1}: $2\int |f-x|$

Variable area
- $c=\frac{a+b}{2}$ for least
- $A(k)$, then $A'(k)=0$
- $f=A'(x)$

Average, RMS
- $\frac{1}{b-a}\int f$
- $\sqrt{\frac{1}{b-a}\int f^2}$

Parametric
- $\int y\,x'(t)\,dt$
- Change limits to t